

FASTQ and FASTQC

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Overview

① FASTQ

② FASTQC

- FASTQ format stores raw sequence data from high-throughput sequencers.
- Each read consists of four lines:
 - ① Identifier: Unique read identifier (e.g., @SEQ_ID)
 - ② Sequence: Nucleotide sequence (e.g., AGCTGAC)
 - ③ Plus Line: A + sign, sometimes with the same identifier (e.g., +SEQ_ID)
 - ④ Quality Scores: ASCII-encoded Phred quality scores

FASTQ File Example

Example

```
@M00970:337:000000-BR5KF:1:1102:17745:1557 1:N:0
CGTATGCCGTCTTCTGCTTGAAAAAAAAAAAAAAAAAAAAACAAA
+
GGFGGGGGGGGGFGGGGGGGGGGGGGGGGGFGGG+38+35*311*6,,3
```

FASTQ Format Description

- @<sequence identifier and metadata>
 - e.g., sequencing machine, flowcell ID, etc.
- <sequence>
 - e.g., CGTATGCCGTCTTCTGCTTG...
- +<optional quality identifier>
 - often just '+' or the sequence identifier repeated
- <quality scores>
 - ASCII-encoded per-base quality scores

Phred Quality Score

- Quality Score (Q) = $-10 \times \log_{10}(P)$, where P is the probability of a base being wrongly sequenced.
- **ASCII encoding** of quality scores: Each quality score is represented by an ASCII character.
 - **Phred quality score (Q)**: The ASCII value of the quality score = Sequence quality + 33.

Quality Score Range:

- Worst Quality: !
(ASCII 33)
- Best Quality: ~
(ASCII 126)

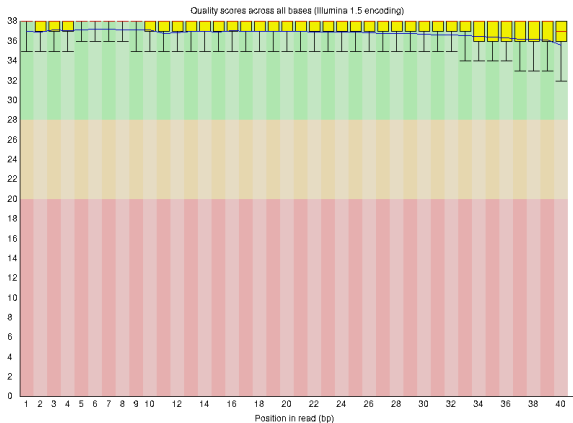
Is the quality of my sequenced data OK?

- After sequencing, it's crucial to assess the **quality** of the data.
- Common indicators of poor quality include:
 - Low average quality scores
 - Adapter contamination
 - GC bias
 - Overrepresented sequences
 - Read length variations
- **Next step:** Tools like **FastQC** are used to evaluate data quality before trimming and downstream analysis.

- **FastQC** is a tool for quality control of high-throughput sequencing data.
- It provides a detailed analysis of the following quality issues:
 - Adapter contamination
 - End-of-read quality drop-off
 - Atypical GC content patterns
 - Overrepresented sequences
 - Other sequencing artifacts
- **Output:** Graphs and visual summaries of quality metrics.
- **Next steps:** Trim, filter, or correct issues before further analysis.
- For more information, visit: [FastQC Official Website](#)

FASTQC: Per Base Sequence Quality

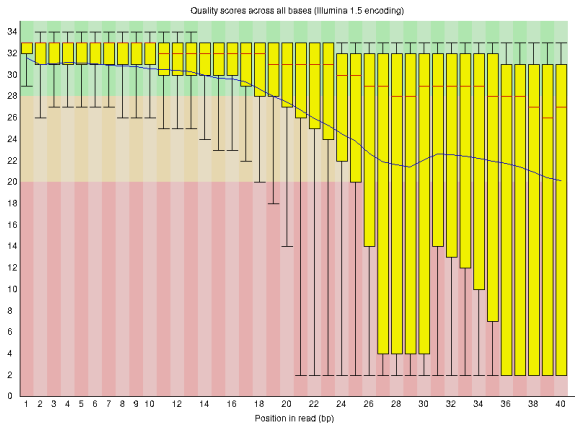
Good Data



- Consistent

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Bad Data



- High Variance

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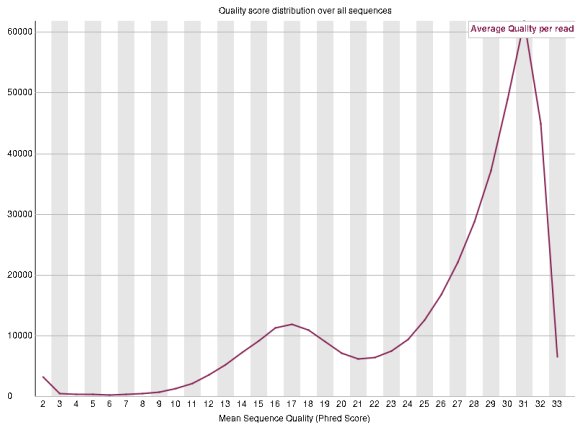
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FASTQC: Per Base Sequence Quality

Good Data



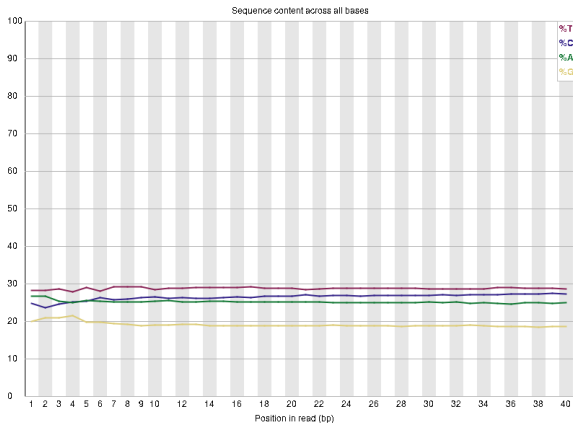
Bad Data



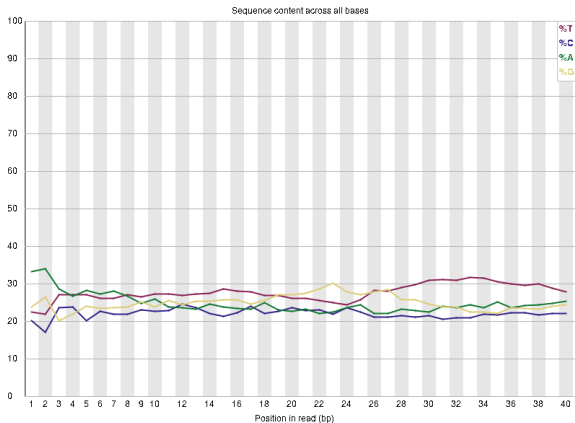
- Non-Uniform Distribution

FASTQC: Per Base Sequence Quality

Good Data



Bad Data



- Smooth Over Length

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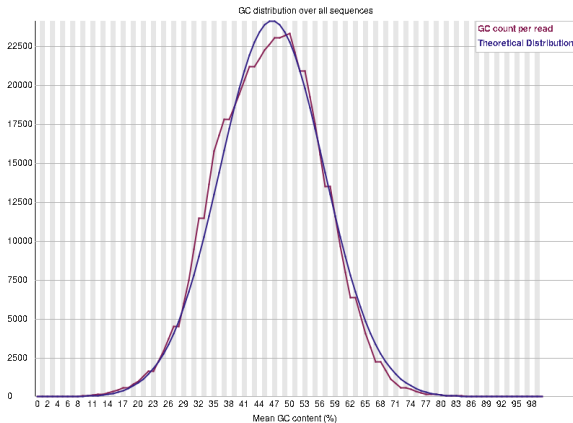
- Sequence-position Bias

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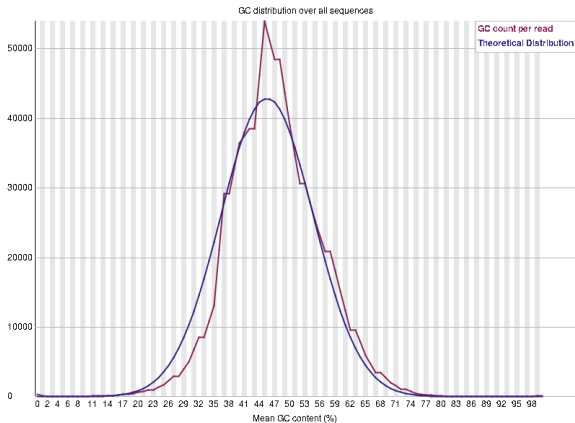
FASTQC: Per Base Sequence Quality

Good Data



- Fits with Expectation

Bad Data



- Does Not Fit with Expectation

References



Harvard Slides

Link: <https://m.youtube.com/watch?v=T2QAGtuvq8I&t=1s&pp=2AEBkAIB>

Harvard University



Harvard Reports

Link: <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>

Harvard University

The End

Questions? Comments?